# Chapter 7 Study Guide: The Water Cycle and Watersheds

Earth Science, Ms. Reyes. Exam Friday, November 6: 40 questions.

Ms. Reyes said to know four things: the terms, the order of the cycle, what a watershed is, and the freshwater numbers. This guide follows that order.

## 1. Where Earth's water is

| Pool | Share |
|---|---|
| Salt water (oceans) | about 97% of all water |
| Fresh water | about 2.5% of all water |

Of the fresh water:

| Where | Share of fresh water |
|---|---|
| Glaciers and ice caps | about 69% |
| Groundwater | about 30% |
| Surface water (lakes, rivers, swamps) | less than 1% |

Only a tiny share of fresh water is easy for people to use.

**Trap:** fresh water is about 2.5% of Earth's water, not "most" of it. The 69 / 30 / 1 split is a share of the fresh water, not of all water.

## 2. The cycle, in order

1. **Evaporation:** liquid water becomes vapor, driven by the sun's heat. It happens at any temperature, not only at boiling.
2. **Transpiration:** plants release water vapor through their leaves.
3. **Condensation:** water vapor cools and becomes liquid droplets. This forms clouds.
4. **Precipitation:** water falls as rain, snow, sleet, or hail.
5. **Runoff:** water flows over land into streams and rivers.
6. **Infiltration:** water soaks into the soil and can recharge groundwater.

Water can follow many paths, so the cycle is not one fixed line. It also has no starting point; it does not "start at the ocean."

**Evaporation vs. transpiration:** evaporation comes from surfaces (oceans, lakes, soil). Transpiration comes from living plants.

## 3. Residence time

Residence time is how long a water molecule stays in one place.

| Reservoir | Residence time |
|---|---|
| Atmosphere | about 9 days |
| Rivers | about 2 weeks |
| Ocean | about 3,200 years |
| Glaciers and ice caps | can be thousands of years |

Memory hook: the fast, moving pools (air, rivers) turn over in days or weeks. The big storage pools (ocean, ice) turn over in thousands of years.

# 4. Groundwater words

- **Aquifer:** a layer of rock or sediment that holds and moves groundwater.
- **Water table:** the top of the zone where the ground is completely saturated. It rises and falls. It is **not flat**; it follows the land a little.
- **Recharge:** water added to an aquifer by infiltration.

# 5. Watersheds

A **watershed** is the land area that drains to one outlet, such as a river, lake, or the sea. Its boundary is a ridge or high ground called a **divide**.

Everything that happens in a watershed (fertilizer, pollution, paving) can affect the water at the outlet.

**Paving:** paved surfaces cut infiltration and increase runoff, so streams can rise faster after storms.

**Water budget from class:**

> Runoff = Precipitation - Infiltration - Evaporation

This ignores plants and storage.

Worked example: 50 mm of rain falls, 20 mm infiltrates, 10 mm evaporates. Runoff = 50 - 20 - 10 = 20 mm. If paving cuts infiltration to 5 mm, runoff = 50 - 5 - 10 = 35 mm. (These numbers are made up for practice, not from your notes.)

# 6. Common mistakes Ms. Reyes warned about

- Mixing up transpiration (plants) and evaporation (surfaces).
- Saying the water table is flat.
- Thinking the water cycle starts at the ocean. It has no starting point.
- Forgetting that fresh water is only about 2.5%, not "most" of Earth's water.

# 7. Self-test

Cover the answers and try these.

1. What percent of Earth's water is fresh? *(about 2.5%)*
2. Where is most fresh water stored? *(glaciers and ice caps, about 69%)*
3. What is the second-largest fresh water pool? *(groundwater, about 30%)*
4. Name the six stages in order. *(evaporation, transpiration, condensation, precipitation, runoff, infiltration)*
5. Which stage forms clouds? *(condensation)*

6. Which stage comes from living plants? *(transpiration)*
7. What is residence time? *(how long a water molecule stays in one place)*
8. Which has the longer residence time, the atmosphere or the ocean? *(ocean, about 3,200 years vs. about 9 days)*
9. What is an aquifer? *(rock or sediment layer that holds and moves groundwater)*
10. Is the water table flat? *(no; it rises and falls and follows the land a little)*
11. What is recharge? *(water added to an aquifer by infiltration)*
12. What is a watershed, and what marks its edge? *(land area draining to one outlet; a divide)*
13. Why do streams rise faster after storms in a paved area? *(less infiltration, more runoff)*
14. Write the water budget. *(Runoff = Precipitation - Infiltration - Evaporation)*
15. Does the water cycle start at the ocean? *(no; it has no starting point)*